

Direct effects of habitat area on interaction diversity in pollination webs

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Abstract. Island biogeography theory predicts that species richness increases with habitat area and declines with isolation. We expand this framework to address changes in the number of links and species in pollination webs from 12 isolated hills, ranging in area from tens to thousands of hectares, immersed in the agriculture matrix of the Argentine Pampas. We also studied whether total interaction frequency is partitioned more evenly among individual links in richer webs. Our results reveal a direct effect of area on the number of links and species present in each pollination web. However, link richness increased twofold faster than species richness with area. These area effects were not confounded by sampling effort or correlated incidence of exotic species, despite widespread habitat disturbance. Habitat proximity, an inverse measure of isolation, had a marginally significant influence on link but not on species richness. Increased link number was associated with decreasing dominance by any particular interaction and increasing interaction evenness. Despite the strong area effect, a rich pollination web sampled from a small, protected sierra suggests that simple conservation measures, such as reduced grazing and fire suppression, may effectively preserve much local interaction diversity.

Key words: area effect; Argentina; dominance; evenness; habitat island; interaction diversity; island biogeography; isolation; link diversity; Pampas; pollination webs; species richness.

INTRODUCTION

The theory of island biogeography (MacArthur and Wilson 1967) postulates that an island's species richness results from a dynamic equilibrium between opposing extinction and immigration rates, which depend on island size and isolation, respectively. Specifically, the limited population size that small islands can support should elevate extinction risk, limiting diversity; this process could be accentuated by competitive dominance of a few thriving species. Infrequent immigration and rescue of dwindling populations can also limit species diversity on isolated islands. This simple conceptual model has been invoked widely to explain patterns of variation in species richness among oceanic and habitat islands differing in either size or isolation (Vuilleumier 1970, Brown 1971, Brown and Dinsmore 1988, Lom-

lino et al. 1989, Nores 1995). This model has also been applied in conservation, particularly in studies of the consequences of habitat fragmentation (Simberloff and Gotelli 1984) and the design of natural reserves (Higgs 1981). Here we expanded this classical theory to address some of the causes of diversity changes in interaction networks.

Long-term population persistence may ultimately relate to the number, nature (i.e., antagonistic or mutualistic), and intensity of interactions among species and the topology of the resulting interaction network (Bascompte et al. 2006, Bastolla et al. 2009). Specifically, in mutualistic plant–animal webs, an increase in the total number of links can increase web resilience, buffering individual species against disruption of any particular interaction (Okuyama and Holland 2008). As populations decline with decreasing habitat area and become more rarely “rescued” with increasing isolation, encounter probability between mutualists should decrease, whereas morphological and phenological mismatch should increase (Mommott et al. 2007, Stang et al.

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2007). In addition, a few thriving species on more depauperate islands may concentrate interaction strength (Bascompte et al. 2006, Aizen et al. 2008), resulting in a more uneven partitioning of total interaction frequency among individual links, which exacerbates the disruption probability of the most vulnerable interactions (Tylianakis et al. 2007). Therefore, either habitat area or isolation should affect the number of interaction links more strongly than species richness; we test this expectation in this study. In the context of habitat fragmentation, this effect should translate into a higher rate of interaction than of species extinctions as habitat remnants shrink and become increasingly isolated.

Although the direction of the effects of habitat area and isolation on number of species and links is straightforwardly predicted by an expanded theory of island biogeography, the relative magnitude of these two predictors is not. Particularly, in pollination webs the magnitude of these effects depends on a multiplicity of factors including pollinator vagility and diversity, abundance and distribution of nesting sites, and floral resources in the target habitat and surrounding matrix (Lonsdorf et al. 2009). More generally, the relative impact of habitat area and isolation on web diversity would also be influenced by the scale and range of these two landscape variables, which are system specific.

We studied whether the number of interactions in pollination webs increases with habitat area and proximity to similar habitats more rapidly than expected from changes in the number of interacting species, and whether total interaction frequency is partitioned more evenly among individual links in richer webs. This study considered plant–pollinator communities on a series of ancient (lower Paleozoic) eroded hills, “sierras,” jutting out of a more modern (Quaternary) matrix, the loessic Pampean plains. These sierras range in area from tens to thousands of hectares and differ in isolation from a few hundred meters to several kilometers. Most of them remain untitled because of steep slopes, shallow soils, and exposed bedrock, and thus they are islands of biodiversity immersed in one of the most intensively used agricultural matrices in South America. Despite being highly disturbed due to cattle grazing, fire, herbicide and pesticide misuse, and species invasion, these sierras still preserve many floristic elements that used to be common in the surrounding plains and elsewhere in southern South America, including endemic species of Gondwanan origin (de la Sota 1967). The biological diversity and conservation interest of these hills make this orographic system ideal for studying concepts of island biogeography.

Studies of island biogeography, especially those of terrestrial habitat islands, are often complicated by various problems, which we considered during sampling design and statistical analysis. To counteract the confounding influences of covarying factors, such as habitat heterogeneity and sampling effort (Stevens

1986), we adopted a sampling protocol and analysis procedure to isolate the direct effects of habitat area on the numbers of species and interaction links in the pollination webs. Another source of confounding factors arises because, unlike oceanic islands, habitat islands are not truly isolated (Janzen 1983), so that organisms, perhaps of alien origin, thriving in the surrounding matrix can cross habitat edges and integrate into focal interaction webs. Depending on the strength and covariation of invasions with different landscape factors, they may eventually negate or exacerbate effects of habitat area or isolation on the numbers of species or interaction links. Although we focus principally on quantitative, rather than qualitative (e.g., whether interactions involve native or alien mutualists) effects of habitat size and isolation on link diversity, we did address whether any relation of invasion to either habitat area or isolation affects our interpretations. Finally, given the direct implications of this study for conservation, we explored whether suppression of severe anthropogenic disturbance could effectively offset direct effects of habitat area on the number of interaction links.

MATERIALS AND METHODS

Study system

The sierras that we surveyed are part of the Tandilian orographic system located to the southeast of the “Pampeano Austral” biogeographic district (Cabrera and Willink 1973). The study area comprises about 24 isolated hills, with maximum altitudes between 400 and 500 m. Climate in the region is temperate with warm summers (mean January temperature 20.8°C) and mild winters (mean July temperature 5.2°C). Typically, most of the sierras include three main sub-habitats: a basal rocky gentle slope dominated by diverse shrubs, herbs, and geophytes; a scarcely vegetated steep scarp; and a flat top with a mosaic of exposed bedrock and patches of loess dominated by grasses. The surrounding matrix, originally an extensive grassland, is dominated by agriculture fields cultivated with soybean, sunflower, wheat, corn, potato, rape, and other crops. Apiculture with European varieties of *Apis mellifera*, the common honey bee, is also an important economic activity in areas surrounding the sierras. However, the sierras support a higher abundance and richness of flower visitors than the surrounding matrix (A. Saez, unpublished data).

For this study, we sampled the pollination webs from 12 sierras located between Mar del Plata (37°58' S, 57°35' W) and Balcarce (37°50' S, 58°15' W). The areas of the study sierras ranged between 13 and 2100 ha and their distances from the nearest sierra ranged between 0.80 and 9.1 km (Appendix A). These values are representative of the range in area and isolation of the whole orographic system. Cattle roamed freely and fires were set frequently to promote regrowth in all the study sierras except for Difuntito, a small hill that was

protected by being completely enclosed within a fenced agricultural field. Cattle grazing and fire had been suppressed at Difuntito for at least the preceding 18 years (O. Tapico, *personal communication*).

Field sampling

Fieldwork was conducted during the 2007–2008 flowering season (October–April). At each study sierra, we set a sampling area of ~0.5 ha each on a north-facing rocky slope, about 200 m from the edge of the nearest agriculture field. This sun-exposed, heterogeneous terrain presents the highest plant diversity among all the sierra sub-habitats. Within each sampling area, we set 10 permanent 1 m radius plots about 25 m apart along two parallel 100-m transects separated by 50 m. After identifying all blooming angiosperm species within each plot, we identified and counted all flower visitors that contacted floral sexual organs during a 15-min period. All plots from a given sierra were censused consecutively between 09:00 and 18:00 hours. We conducted observations in two sierras per day, and each sierra was sampled 10 times, on average, throughout the flowering season, once every two weeks. In total, we accumulated 318 h of observation distributed over 127 days. Only visited plant species were included in the analysis. Individual animals observed visiting flowers were morphotyped and identified to the minimum possible taxonomic level with the aid of a reference collection and the expertise of different specialists. Flower visitors that could be identified to species, or at least to genus, accounted for 52% and 80% of all flower visitors, respectively. Hereafter, we refer to all morphospecies as species.

Based on plant and insect origin (alien vs. native), we estimated the proportion of all flower visitors visiting flowers of alien species and the proportion of all flower visitors known as alien (Appendix B). The average of these proportions was considered as an index of invasion of a pollination web, which ranges from 0 for a web with no interacting alien species to 1 for a web characterized exclusively by interactions between alien species (Aizen et al. 2008). We also estimated the proportion of all flower visits by *Apis mellifera* (Appendix B), the dominant alien visitor.

Data analysis

Using multiple linear regression, we evaluated the effects of habitat area and proximity, an inverse measure of isolation, on the numbers of interaction links, species in the web (i.e., plants plus flower visitors), and observed total number of flower visitors (as a measure of sampling effort and total interaction frequency). We conducted this analysis considering these variables for the complete web and the native sub-web (i.e., number of links between native species of plants and flower visitors, number of native species, and total number of native flower visitors recorded on flowers of native species). All independent and dependent variables were log-transformed, both to increase normality of residuals and to

facilitate interpretation of regression coefficients in proportional terms (Gelman and Hill 2007). We used a similar analysis to test whether the (log-transformed) number of alien species and (arcsine square-root transformed) invasion index and proportion of honeybees were related to area and habitat proximity. All regression analyses were carried out excluding Difuntito, the small protected sierra, from the database.

Among many different indices of habitat proximity (reviewed in Winfree et al. 2005), we estimated habitat proximity for sierra j by calculating

$$\sum_{i \neq j} A_i \exp(-\beta d_{i,j}) \quad (1)$$

where A_i is the area of sierra i in hectares and $d_{i,j}$ is the minimum distance (km) between the edges of sierra i and j , including all the sierras i . Adjacent sierras were included in this index if they were within 10 km, the longest foraging distance for bees (Steffan-Dewenter et al. 2001, Gathmann and Tschardt 2002, Steffan-Dewenter and Tschardt 2002), of a focal sierra (j). β was set to 0.46 following the results of the Ricketts et al. (2008) meta-analysis of edge effects on bee abundance and diversity. This coefficient results in 50% decay in bee species richness with isolation of 1.5 km. The use of other β values (e.g., $\beta = 1$; Steffan-Dewenter and Tschardt 2000) or simpler empirically derived indexes, such as the proportion of the total area occupied by sierra habitat within 1.6 km (Greenleaf et al. 2007) or 2.4 km (Kremen et al. 2004) from a given focal sierra, did not modify our statistical conclusions.

For each sierra, we characterized the partition of total interaction frequency among individual links with dominance and evenness indices. Here, for each individual link, interaction frequency was estimated as the total number of flower visitors from a given species to the flowers of a particular plant species over the entire flower season (Olesen et al. 2002). Dominance (D) was then estimated as the most frequently observed interaction between a particular plant and flower-visiting species, expressed as a proportion of the total interaction frequency (Berger and Parker 1970). Because this simple dominance index can be affected by the number of links in the web (L), we corrected it by subtracting $1/L$, the expected value of D when total interaction frequency is equally partitioned among links. To characterize evenness, we used Hulbert's PIE index (probability of an interspecific encounter; Hurlbert 1971), which here is related to the chance that two randomly selected flower visitors either belong to different species or visit flowers from plants of different species. This index approaches 0 for a single dominant interaction and reaches 1 when total interaction frequency is equally partitioned among links independent of the number of links. The association between dominance or evenness and link number was assessed using Spearman's rank correlation.

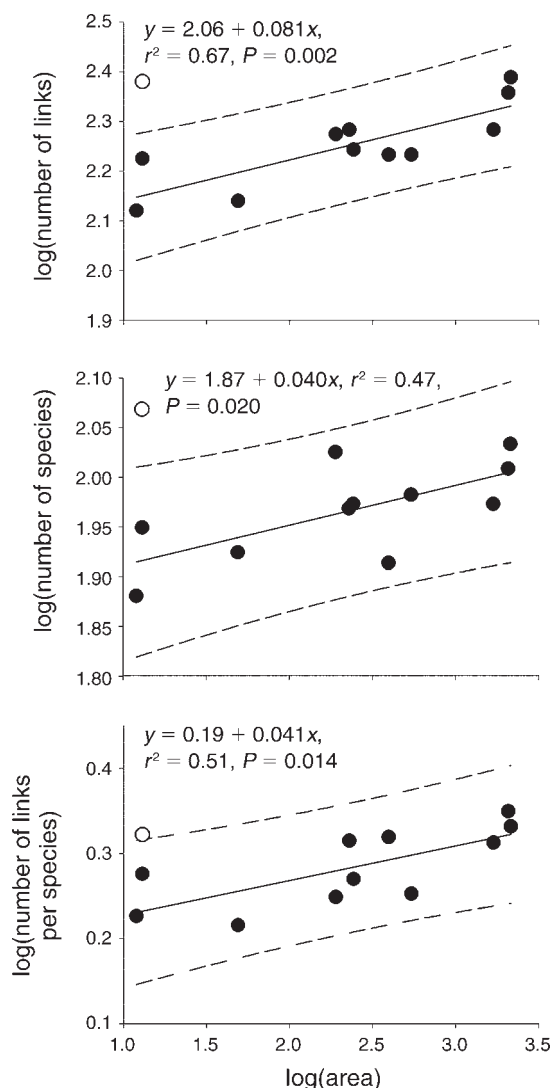


FIG. 1. Relationships of the total number of interaction links, number of species (plants + animals), and links or species ratio to area (originally measured in ha) of the sierra for 11 pollination webs in the Argentine Pampas. Each solid circle represents one of the 11 webs. Plots correspond to simple linear regressions after excluding the nonsignificant effect of habitat proximity. Dashed curves indicate the boundaries of the 95% prediction interval. The open circle in each panel represents the estimate for the well-preserved web sampled at Difuntito, which was excluded from calculations. All variables were $\log_{10}$ -transformed.

RESULTS

The sizes (76–117 species) and numbers of interactions (132–245 links) of the study networks characterize them as diverse, particularly when compared with other local pollination webs from temperate regions (Jordano 1987, Aizen et al. 2008) and oceanic islands (Olesen et al. 2002). We recorded 13 174 flower visitors from 171 species, which were involved in 1168 different interactions with 96 plant species. All flower visitors were

insects, except for the hummingbird *Chlorostibon aureoventris*. The assemblage of flower visitors was dominated by Hymenoptera (47% and 44% of all flower visitors and species, respectively), followed by Diptera (24% and 21%), Coleoptera (22% and 20%), and Lepidoptera (5% and 14%). The other orders, Hemiptera and Aves, were represented by <1% of all visitors and species. The most frequent flower visitor was the alien honey bee *Apis mellifera*, accounting for ~20% of observed visitors. Among plants, the most visited and diverse plant families were Asteraceae (50% and 41% of all flower visitors and species, respectively), followed by Apiaceae (31% and 6%), Fabaceae (5% and 8%) and Scrophulariaceae (4% and 1%). The three most-visited species, *Eryngium stenophyllum* (24%), *Bacharis tandilensis* (7%), and *Achyrocline satureioides* (5%), were all native.

The numbers of interaction links and species present in each sampled pollination web varied positively with sierra size (Fig. 1). The partial regression coefficient for the effect of habitat proximity on the number of links was positive, as expected, but only marginally significant ($P = 0.098$). Effects of area and habitat proximity on the number of links between native species and total number of native species visiting native flowers paralleled the patterns for the complete webs (Appendix C). The lack of strong effects of habitat proximity could not be attributed to a confounding effect of area, as these two (log-transformed) predictors correlated weakly ($r = 0.41$, $n = 11$, $P = 0.21$) and the associated variance inflation factor of 1.20 is much smaller than the value of 10, which indicates strong multicollinearity (Quinn and Keough 2002). The area of a sierra by itself explained a significant and large percentage of the variation in both the number of links (67%) and species (47%) in each web. However, the number of links increased faster with sierra size than did the number of species (Fig. 1); an area increase of one order of magnitude related to an average ~20% increase in link number, but of only ~10% increase in species richness. Thus, the ratio of numbers of links to species also increased with area (Fig. 1). Indeed, area positively affected link number even after accounting for variation in the (log-transformed) number of species among webs (partial regression coefficient $\pm$ SE = 0.043 ± 0.019 , $t_8 = 2.21$, $P = 0.058$). In the context of this area effect, the network in the only well-preserved sierra, Difuntito, was clearly exceptional in terms of species richness, link number, and even the number of links per species, each of which exceeded the upper limits of the 95% prediction intervals estimated from the regressions based on the other 11 sierras (Fig. 1).

Area effects on the numbers of links and species could not be attributed to confounding study factors. The total numbers of individual flower visitors or native visitors visiting native flowers (measures of sampling effort) did not clearly vary with either area or habitat proximity (Appendix C). Also, although some webs were highly

invaded (e.g., Piedra Alta; Appendix B), the extent of invasion as measured by the number of alien species, the invasion index, and the proportion of honey bees, did not vary significantly with either of the two landscape predictors (Appendix D). Thus, sierra size seems to affect the numbers of species and interaction links independently of other study factors.

The partitioning of interaction frequency among links became increasingly equal as link number increased. Dominance correlated negatively with the number of interaction links in the web, even though the identity of the most frequent interaction varied among sierras (Appendix B). Furthermore, the PIE index, a more direct measure of evenness, increased with link number (Fig. 2). Similar results were observed when the analysis considered only the hierarchy of interaction links between native species ($r_s = -0.66$, $P = 0.024$ for dominance; and $r_s = 0.74$, $P = 0.008$ for evenness). As a consequence, even the weakest interaction link of a relatively rich pollination network was comparatively stronger than that of a relatively poor network, for both the whole web and the native sub-web.

DISCUSSION

In the study system, sierra size affected the number of links and species independently of possible artifacts such as sampling effort, the area surveyed, or the sampling time (see also Stevens 1986). Other potentially confounding factors did not play any major role. For instance, although alien species such as *Apis mellifera* and some agricultural weeds are important components of some of the studied pollination webs, their incidence (as either the number of species or their fraction of the total interaction frequency) was unrelated to sierra area or to proximity to other sierras. In fact, the area effect basically reflected changes in the number of links between native plants and animals and the total number of native species, which include many endemics. Finally, the sierra size effect occurred under moderate to severe anthropogenic disturbance, including browsing and trampling by cattle and frequent burning, suggesting that large, continuous expanses of even highly modified natural habitats increase both the number of species they harbor and the number of interaction links present in mutualistic webs.

Results supported our expectation that the number of links in a pollination web increases more rapidly with fragment size than does the number of species. For instance, the web sampled in the largest sierra, Vigilancia, displayed 86% more links, but only 42% more species than the web sampled in the smallest sierra, La Paja. This finding could also indicate a faster loss of interaction links than of species as isolated habitats become reduced. This differential loss occurs according to the following empirical rule of thumb: the disruption of two interaction links involves, on average, the extirpation of one species in a pollination web. Thus, the disappearance of interactions from a pollination web

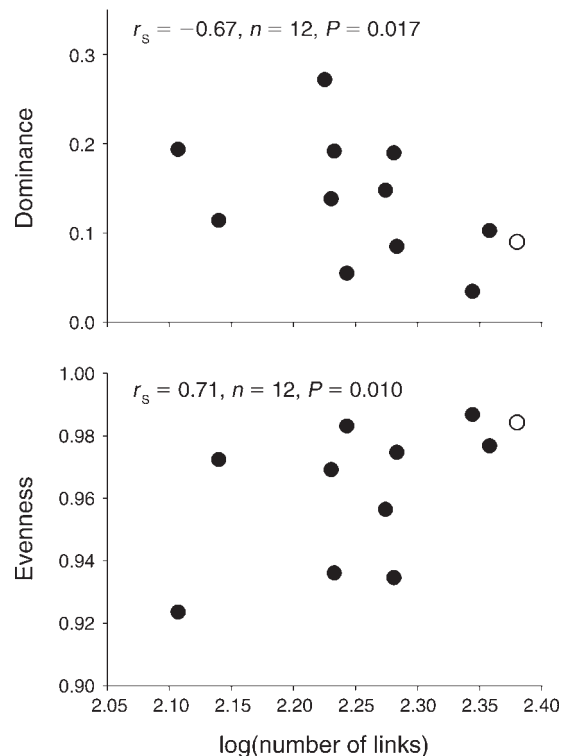


FIG. 2. Relationships of interaction dominance and evenness, estimated by the PIE index (probability of an interspecific encounter; Hurlbert 1971), to the number of links for 12 pollination webs. Each solid circle represents one of the 11 webs. The open circle represents the web sampled at Difuntito, the well-preserved small sierra (protected from fire and grazing).

might anticipate the loss of interacting species, supporting the proposal that connectivity in mutualistic webs increases stability (Okuyama and Holland 2008). This finding has a direct applied implication: namely, that decreases in interaction numbers can provide clues to proximate local species extinctions (Aizen and Feinsinger 1994, Tylianakis et al. 2007).

We also found support for the proposed mechanism to explain a faster increase, with sierra size, in the number of links than in the number of species. Basically, a more equal distribution of total interaction frequency among individual plant–pollinator links in richer webs (Fig. 2) probably reflects a higher chance of encounter between mutualist partners associated with more abundant and stable populations in large than in small sierras (Tylianakis et al. 2007). Also, large populations should harbor more phenotypic variation that may reduce the likelihood of morphological and phenological mismatches (Jordano et al. 2003, Stang et al. 2007), which may be particularly critical for the occurrence of interactions between specialists. On the other hand, several small plant and pollinator populations may still persist in small habitat islands, interacting with highly abundant and generalist species, at the same time that

their chance to interact with other rare partners decreases as habitat shrinks and these populations become increasingly smaller (Vázquez and Aizen 2003). Thus, the dynamics of link loss would involve the disruption of the weakest links and persistence of the specialist species through asymmetrical interactions with the more generalist species. We then predict that this differential link disruption would result in increasingly asymmetric and nested network configurations (Bascompte et al. 2003, Vázquez and Aizen 2004), which buffer pollination webs against further species loss (Bascompte et al. 2006, Bastolla et al. 2009).

A marginal effect of sierra proximity on the number of links suggests that web diversity was at best weakly influenced by dispersal from nearby sierras. Any rescue effect related to sierra proximity may be blurred by a stronger effect of the surrounding matrix (Cook et al. 2002) and the movement of flower visitors and plant propagules within metapopulations in each sierra (see Hanski 1999). In any event, the incidence of alien mutualists was very uneven among webs and was unrelated to sierra size or isolation (Appendix D). In particular, the high frequency of *Apis mellifera* in some sierras is probably influenced by the presence of nearby apiaries in the matrix. This external influence from the surrounding matrix, perhaps related to matrix heterogeneity and nearby sierras, could explain some of the turnover in the identity of the dominant interaction from each entire web or native sub-web across the different sierras (Appendix B). Indeed, a partial shift in the compositional makeup and hierarchy of existing interactions over space and time is congruent with an immigration–extinction dynamic equilibrium in the number of interaction links expected from an expanded theory of island biogeography, although of course other factors such as disturbance history are important.

Human disturbance can affect the diversity and architecture of mutualistic webs (Vázquez and Simberloff 2003, Pauw 2007, Aizen et al. 2008). Despite the observed strong area effect on the number of interaction links, suppression of roaming cattle and fire might greatly increase diversity of pollination webs, even in small sierras (see also Aschero and Vázquez 2009). Regrettably, species and interactions from the study habitats, part of the last few remnants supporting Pampean biodiversity, are not acknowledged as warranting conservation status, and most sierras in the Tandilia system are subject to frequent human disturbance. There is some room for hope, however, if conservation actions were to be undertaken. The small protected sierra, Difuntito, provides evidence that simple conservation measures, such as fencing and suppression of sustained anthropogenic disturbance, can have large biodiversity consequences for pollination webs, despite the incidence of some agricultural weeds and honey bees. Indeed, this small sierra harbors a pollination web that, at least at the local scale, was as

rich as any other web sampled in sierras over two orders of magnitude larger. More replicates of pollination webs from protected sierras, which are unavailable in our study system, are needed to establish a conclusive benefit of such measures. However, the unexpected diversity of this pollination web suggests that anthropogenic disturbance may interact with area, and eventually with other landscape features, to set the biodiversity contained in mutualistic webs from habitat fragments (Aizen 2007). Further comparisons of even simple network descriptors, such as number of species and links, along different landscape gradients can provide relevant information on web dynamics with implications for conservation.

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LITERATURE CITED

- Aizen, M. A. 2007. Enfoques en el estudio de la reproducción sexual de las plantas en ambientes alterados: limitaciones y perspectivas. *Ecología Austral* 17:7–19.
- Aizen, M. A., and P. Feinsinger. 1994. Forest fragmentation, pollination, and plant reproduction in a Chaco dry forest, Argentina. *Ecology* 75:330–351.
- Aizen, M. A., C. L. Morales, and J. M. Morales. 2008. Invasive mutualists erode native pollination webs. *PLoS Biology* 6(2): e31. [doi: 10.1371/journal.pbio.0060031]
- Aschero, V., and D. P. Vázquez. 2009. Habitat protection, cattle grazing and density-dependent reproduction in a desert tree. *Austral Ecology* 34:901–907.
- Bascompte, J., P. Jordano, C. J. Melián, and J. M. Olesen. 2003. The nested assembly of plant–animal mutualistic networks. *Proceedings of the National Academy of Sciences USA* 100:9383–9387.
- Bascompte, J., P. Jordano, and J. M. Olesen. 2006. Asymmetric coevolutionary networks facilitate biodiversity maintenance. *Science* 312:431–433.
- Bastolla, U., M. A. Fortuna, A. Pascual-García, A. Ferrera, B. Luque, and J. Bascompte. 2009. The architecture of mutualistic networks minimizes competition and increases biodiversity. *Nature* 458:1018–1020.
- Berger, W. H., and F. L. Parker. 1970. Diversity of planktonic foraminifera in deep-sea sediments. *Science* 168:1345–1347.
- Brown, J. H. 1971. Mammals on mountaintops: nonequilibrium insular biogeography. *American Naturalist* 105:467–478.
- Brown, M., and J. J. Dinsmore. 1988. Habitat islands and the equilibrium theory of island biogeography: testing some predictions. *Oecologia* 75:426–429.
- Cabrera, A. L., and A. Willink. 1973. Biogeografía de América latina. Organización de Estados Americanos, Washington, D.C., USA.
- Cook, W. M., K. T. Lane, B. L. Foster, and R. D. Holt. 2002. Island theory, matrix effects and species richness patterns in habitat fragments. *Ecology Letters* 5:619–623.
- de la Sota, E. R. 1967. Composición, origen y vinculaciones de la flora pteridológica de las sierras de Buenos Aires

- (Argentina). *Boletín de la Sociedad Argentina de Botánica* 11:105–128.
- Gathmann, A., and T. Tschardt. 2002. Foraging ranges of solitary bees. *Journal of Animal Ecology* 71:757–764.
- Gelman, A., and J. Hill. 2007. *Data analysis using regression and multilevel/hierarchical models*. Cambridge University Press, Cambridge, UK.
- Greenleaf, S. S., N. M. Williams, R. Winfree, and C. Kremen. 2007. Bee foraging ranges and their relationship to body size. *Oecologia* 153:589–596.
- Hanski, I. 1999. Habitat connectivity, habitat continuity, and metapopulations in dynamic landscapes. *Oikos* 87:209–219.
- Higgs, A. J. 1981. Island biogeography theory and nature reserve design. *Journal of Biogeography* 8:117–124.
- Hurlbert, S. H. 1971. The nonconcept of species diversity: a critique and alternative parameters. *Ecology* 52:577–586.
- Janzen, D. H. 1983. No park is an island: increase in interference from outside as park size decreases. *Oikos* 41:402–410.
- Jordano, P. 1987. Patterns of mutualistic interactions in pollination and seed dispersal: connectance, dependence asymmetries, and coevolution. *American Naturalist* 129:657–677.
- Jordano, P., J. Bascompte, and J. M. Olesen. 2003. Invariant properties in coevolutionary networks of plant–animal interactions. *Ecology Letters* 6:69–81.
- Kremen, C., N. M. Williams, R. L. Bugg, J. P. Fay, and R. W. Thorp. 2004. The area requirements of an ecosystem service: crop pollination by native bee communities in California. *Ecology Letters* 7:1109–1119.
- Lomolino, M. V., J. H. Brown, and R. Davis. 1989. Island biogeography of montane forest mammals in the American Southwest. *Ecology* 70:180–194.
- Lonsdorf, E., C. Kremen, T. Ricketts, R. Winfree, N. Williams, and S. Greenleaf. 2009. Modelling pollination services across agricultural landscapes. *Annals of Botany* 103:1589–1600.
- MacArthur, R. H., and E. O. Wilson. 1967. *The theory of island biogeography*. Princeton University Press, Princeton, New Jersey, USA.
- Memmott, J., P. G. Craze, N. M. Waser, and M. V. Price. 2007. Global warming and the disruption of plant–pollinator interactions. *Ecology Letters* 10:710–717.
- Nores, M. 1995. Insular biogeography of birds on mountain tops in north-western Argentina. *Journal of Biogeography* 22:61–70.
- Okuyama, T., and J. N. Holland. 2008. Network structural properties mediate the stability of mutualistic communities. *Ecology Letters* 11:208–216.
- Olesen, J. M., L. I. Eskildsen, and S. Venkatasamy. 2002. Invasion of pollination networks on oceanic islands: importance of invader complexes and endemic super generalists. *Diversity and Distributions* 8:181–192.
- Pauw, A. 2007. Collapse of a pollination web in small conservation areas. *Ecology* 88:1759–1769.
- Quinn, G. P., and M. J. Keough. 2002. *Experimental design and data analysis for biologists*. Cambridge University Press, Cambridge, UK.
- Ricketts, T. H., et al. 2008. Landscape effects on crop pollination services: are there general patterns? *Ecology Letters* 11:499–515.
- Simberloff, D., and N. Gotelli. 1984. Effects of insularization on plant-species richness in the prairie–forest ecotone. *Biological Conservation* 29:27–46.
- Stang, M., P. G. L. Klinkhamer, and E. van der Meijden. 2007. Asymmetric specialization and extinction risk in plant–flower visitor webs: a matter of morphology or abundance? *Oecologia* 151:442–453.
- Steffan-Dewenter, I., U. Munzenberg, and T. Tschardt. 2001. Pollination, seed set and seed predation on a landscape scale. *Proceedings of the Royal Society B* 268:1685–1690.
- Steffan-Dewenter, I., and T. Tschardt. 2000. Butterfly community structure in fragmented habitats. *Ecology Letters* 3:449–456.
- Steffan-Dewenter, I., and T. Tschardt. 2002. Insect communities and biotic interactions on fragmented calcareous grasslands: a mini review. *Biological Conservation* 104:275–284.
- Stevens, G. C. 1986. Dissection of the species–area relationship among wood-boring insects and their host plants. *American Naturalist* 128:35–46.
- Tylianakis, J. M., T. Tschardt, and O. T. Lewis. 2007. Habitat modification alters the structure of tropical host–parasitoid food webs. *Nature* 445:202–205.
- Vázquez, D. P., and M. A. Aizen. 2003. Null model analyses of specialization in plant–pollinator interactions. *Ecology* 84:2493–2501.
- Vázquez, D. P., and M. A. Aizen. 2004. Asymmetric specialization: a pervasive feature of plant–pollinator interactions. *Ecology* 85:1251–1257.
- Vázquez, D. P., and D. Simberloff. 2003. Changes in interaction biodiversity induced by an introduced ungulate. *Ecology Letters* 6:1077–1083.
- Vuilleumier, F. 1970. Insular biogeography in continental regions. I. The northern Andes of South America. *American Naturalist* 104:373–378.
- Winfree, R., J. Dushoff, E. E. Crone, C. B. Schultz, R. V. Budny, N. M. Williams, and C. Kremen. 2005. Testing simple indices of habitat proximity. *American Naturalist* 165:707–717.

APPENDIX A

Characteristics of the 12 plant–pollinators webs analyzed (*Ecological Archives* A020-055-A1).

APPENDIX B

Description of the extent of the invasion and the dominant interaction link for the complete webs and among natives (*Ecological Archives* A020-055-A2).

APPENDIX C

Multiple linear regressions of number of links, number of species, and number of flower visitors on the area of the focal sierra and proximity to other sierras for entire webs and native sub-matrices (*Ecological Archives* A020-055-A3).

APPENDIX D

Multiple linear regressions of number of alien species, invasion index, and proportion of honey bees on the area of the focal sierra and proximity to other sierras (*Ecological Archives* A020-055-A4).